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1-A:

The principal-agent problem is an economic concept where one person or entity (the agent) is able to make decisions on behalf of, or that impact, another person or entity (the principal). The problem arises when the agent has different priorities or information than the principal, which can lead to the agent acting in their own self-interest rather than the principal's best interest. In this case, the manager (agent) makes decisions on behalf of the owner (principal), creating a potential conflict of interest.

Why are others options incorrect?

Option B: This scenario describes market speculation or ticket scalping, where an individual buys a product with the intent to resell it for a profit. This is a standard market transaction and does not involve an agent acting on behalf of a principal.

Option C: This scenario describes a monopsony, which is a market condition where there is only one buyer for a product or service. This is a type of market structure, not a principal-agent relationship conflict.

Option D: This scenario describes a monopoly, which is a market condition where there is only one seller of a product or service. This is also a type of market structure, not related to the principal-agent problem.

2-A:

Point Y is lower on the long-run average cost (LRAC) curve than point X, indicating the firm is achieving lower average costs per unit of output. This movement down the LRAC curve is due to economies of scale, which are cost advantages that enterprises obtain due to size, output, or scale of operation.

Option A describes a bulk-buying economy of scale, where a larger firm can negotiate discounts, thus lowering average costs.

Options B, C, and D describe potential causes of diseconomies of scale (e.g., bureaucracy, labor issues), which would cause costs to rise, moving the firm up the LRAC curve or onto a higher curve.

3-C:

In microeconomics, a firm maximises profit where: $MC = MR$

- In the diagram, MC intersects MR around Q_1 .
- Therefore Q_1 is approximately the profit-maximising output.

At this point:

- The difference between AR (price) and AC (cost) is greatest
- This means profit per unit is highest.
 - So total profit is highest near Q_1 .

When firms moving from Q_1 to Q_2 : $MC > MR$

This means the cost of producing the extra unit is greater than the revenue gained from selling it. So the extra unit adds more to cost than to revenue, which reduces profit. Thus: Total profit falls.

Total revenue when output increases and its achieved when $MR=0$

At Q_2 , MR is still positive (above the quantity axis). If $MR > 0 \rightarrow$ total revenue is increasing. So producing more output from Q_1 to Q_2 increases total revenue.

Therefore the correct option is C where total profit is reduced and total revenue is increased.

4-A:

The condition for utility maximization is given by the equal marginal utility per dollar spent for both goods: $MUX/PX = MUY/PY$

The consumer initially satisfies this condition. The goal is to explain what causes the consumer to purchase more of good Y and less of good X. This requires a change that makes the condition : MUY/PY relatively more attractive than MUX/PX at the initial consumption bundle.

(A) an increase in the marginal utility of good Y: An increase in MUY makes the ratio MUY/PY greater than MUX/PX at the initial bundle. The consumer will substitute towards good Y and away from good X to restore the equality, thus buying more Y and less X. This is the correct option

(B) an increase in the price of good Y: An increase in PY makes the ratio MUY/PY smaller than MUX/PX . The consumer will substitute towards good X and away from good Y, buying less Y and more X. This is the opposite of what is required.

(C) an increase in the tax on the consumption of good Y: This is equivalent to an increase in the price of good Y, PY . The effect is the same as option B; the consumer would buy less Y and more X.

(D) an increase in the tax on the income of consumers: This reduces the consumer's income. Since both X and Y are normal goods, a decrease in income leads to a decrease in the demand for both goods. It does not cause an increase in the consumption of Y and a decrease in the consumption of X.

5-D:

A Pareto-optimal outcome, or Pareto efficiency, is a state where it is impossible to make one individual or group better off without making another individual or group worse off.

Options A, B, and C all involve reallocating existing resources or changing production methods in ways that inherently create trade-offs, making one group potentially worse off (e.g., customers not in the loyalty scheme, producers of low-priced goods, or workers in labour-intensive industries).

Option D, training low-skilled workers, represents an improvement in the quality of a resource (labor). This increases overall productivity and efficiency, allowing for more output

for everyone without necessarily making anyone worse off, moving the economy towards a more efficient state.

6-A:

In a free market with negative production externalities, producers only consider their private costs (MPC) and private benefits (MPB), leading to production at the market equilibrium where $MPC = MPB$ (quantity Q_1).

However, the socially optimal level of output occurs where Marginal Social Cost (MSC) equals Marginal Social Benefit (MSB) (quantity Q_2). Since the free market produces Q_1 , which is greater than the socially optimal Q_2 , the result is overproduction from society's perspective.

For all units produced between Q_2 and Q_1 , the marginal social cost is greater than the marginal social benefit, meaning society incurs more costs than benefits. This excess of social costs over benefits results in a deadweight welfare loss, which is represented by the triangle area between the MSC and MSB curves from Q_2 to Q_1 , which is area VWX

7-D:

Revenue Maximisation: A firm aiming to maximise revenue will produce at the output level where Marginal Revenue (MR) is equal to zero ($MR=0$). In the diagram, this quantity corresponds to point Y.

Sales Maximisation: A firm aiming to maximise sales (subject to a normal profit constraint) will produce at the output level where Average Cost (AC) is equal to Average Revenue (AR) (or price), ensuring zero economic profit. In the diagram, this quantity corresponds to point Z.

Effect of Change: When the firm changes its objective from revenue maximisation (quantity X) to sales maximisation (quantity Z), the quantity produced increases from X to Z.

8-D:

A sales tax is a per-unit tax on the good. The initial situation is a 10% sales tax, represented by the supply curve S_1 . The supply curve represents the cost of production plus the tax. When the sales tax is reduced to 5%, the cost for producers at any given quantity decreases. This causes the supply curve to shift downwards (vertically) by the amount of the tax reduction, or equivalently, to shift to the right (horizontally).

Diagram A shows a rightward shift of the demand curve, which is incorrect.

Diagram B shows a leftward shift of the demand curve, which is incorrect.

Diagram C shows a leftward shift of the supply curve, which would represent an increase in tax or costs.

Diagram D shows a rightward shift of the supply curve (from S_1 to S_2), which correctly illustrates a decrease in the cost of supply due to the reduced sales tax. The new equilibrium would be at a lower price and a higher quantity.

9-B:

A negative externality occurs when the production or consumption of a good imposes a cost on a third party. The free market tends to overproduce goods with negative externalities because producers do not account for these external costs.

Option B is the correct policy because imposing a tax on the activity causing the externality (pollution) forces the producer to internalize the external costs. This increases their private costs, reduces the quantity produced to a socially optimal level, and corrects the market failure.

Option A (guaranteed minimum prices) and Option C (rent controls) are forms of price controls (price floor and price ceiling, respectively). These policies distort market prices and quantities but are not designed to address externalities.

Option D (provision of free books) is a policy typically used to address a positive externality (an external benefit, such as improved education and literacy for society) or to address income inequality. Governments use subsidies or direct provision to encourage more production or consumption of goods with positive externalities, which are typically underproduced by the free market.

10-B:

A poverty trap occurs when the marginal net reward for working is very low or negative, creating a disincentive for individuals to seek employment or increase their working hours. Means-tested benefits are payments made to individuals based on their income level. As an individual's earnings increase, their benefits are withdrawn, and they may also have to pay more taxes. This combination can result in little or no overall gain in income, effectively trapping them in poverty.

(A) Legal minimum wage: A legal minimum wage makes work more financially rewarding, which generally helps people increase their income and can assist in escaping poverty, not trapping them.

(C) Prevention of cheaper imports: This is a trade policy designed to protect domestic industries. While it affects the broader economy and employment, it does not create the direct disincentive to earn more income through the benefits system that defines the poverty trap.

(D) Proportional taxation: This tax system takes the same percentage of income from everyone. It does not disproportionately penalize earning more at lower income levels in the same way that the withdrawal of means-tested benefits does.

9-C:

Nudge theory involves influencing behavior through indirect suggestions and positive reinforcement without forbidding any options or significantly changing their economic incentives.

(A) and (D) are mandates or restrictions, not nudges.

(B) is a financial incentive/disincentive, which is a significant change in economic incentives, not a nudge.

(C) aims to inform and subtly encourage voluntary change in behavior (safeguarding fish stocks) through information, which is a classic example of a nudge.

12-A:

A monopsonist hires workers where:

$$MRP_{\square} = MC_{\square}$$

- This occurs at point R.
- Employment = Q_1
- Wage paid = W_1 (from the labour supply curve).

However, the socially efficient level of employment occurs where:

$$MRP_{\square} = S_{\square}$$

- This is point T.
- Employment = Q_3
- Wage = W_3

Because the monopsonist hires Q_1 instead of Q_3 , there is underemployment.

Deadweight loss before NMW

The lost welfare is the triangle between:

- $MRP_{\square}$ (demand)
- Supply of labour
- Quantities Q_1 to Q_3
- This area is RST.
- Initial DWL = triangle RST

2. After the National Minimum Wage

The government sets a minimum wage (NMW) between W_1 and W_3 .

In monopsony:

- The marginal cost of labour becomes horizontal at the NMW.
- The firm now hires where: $MRP_{\square} = NMW$. This happens at point U.

So employment increases from: $Q_1 \rightarrow Q_2$. This is important because minimum wage increases employment in a monopsony.

3. New Deadweight Loss

The efficient level is still Q_3 . But the firm now produces Q_2 instead of Q_1 , so the inefficiency becomes smaller. The remaining DWL is the triangle: U V T

So: New DWL = triangle UVT

4. Final Effect

Deadweight loss changes from: RST → UVT

Since UVT is smaller, the welfare loss falls.

13-D:

With pollution permits, authorities issue a fixed number of permits, which directly caps the total amount of pollutants that can be emitted. This provides certainty about the maximum level of pollution. In contrast, a tax on pollutants incentivizes firms to reduce emissions, but the exact total reduction in pollution is less certain as it depends on how firms respond to the cost (e.g., the elasticity of demand for permits).

The other options are incorrect:

- (a) Firms can typically sell surplus permits in a cap-and-trade system.
- (b) Monitoring is still required to ensure firms do not emit more than their allocated permits.
- (c) Permits aim to achieve an optimal level of pollution, not necessarily zero, which might be prohibitively expensive to achieve.

14-B:

The quantity theory of money posits that there is a direct and proportional relationship between the amount of money in circulation and the general price level of goods and services. In its simplest form, the equation of exchange is given by $MV=PT$ (or $MV=PY$, where Y is real output), where M is the money supply, V is the velocity of money, P is the price level, and T (or Y) is the volume of transactions (or real output). Assuming that the velocity of money (V) and the real output (T or Y) are constant in the short run, any change in the money supply (M) will result in a proportional change in the price level (P).

15-A:

Keynesian economic analysis focuses on aggregate demand and the reasons for recessions and unemployment. A central concept is the "liquidity trap," where monetary policy becomes ineffective because interest rates are already so low that people prefer to hoard cash rather than invest or spend. In this scenario, traditional interest rate cuts fail to stimulate the economy.

B an equilibrium price that always clears the market: This concept is central to classical economics, not Keynesian. Classical economists believe that flexible prices and wages ensure markets are always clear (supply equals demand), leading to full employment in the long run. Keynesians argue that prices and wages can be sticky, leading to persistent unemployment and market disequilibrium.

C a small value for the government expenditure multiplier: The government expenditure multiplier is a key component of Keynesian theory. Keynesians argue that the multiplier is relatively large, meaning a small increase in government spending can have a significant

positive impact on aggregate demand and GDP. A small value for the multiplier would align more with alternative economic theories that emphasize crowding out effects.

D a vertical short-run aggregate supply curve: A vertical short-run aggregate supply (SRAS) curve implies that output is fixed at the full employment level regardless of the price level, which is a classical or monetarist view. Keynesian theory posits a relatively flat or upward-sloping SRAS curve in the short run, suggesting that output can be increased by increasing aggregate demand, especially during a recession when there are unemployed resources.

16-D:

Keynesian economics focuses on aggregate demand. Cyclical unemployment occurs during economic downturns when there is a lack of demand for goods and services. A Keynesian measure to address this involves the government intervening to stimulate demand. This is typically done through expansionary fiscal policy, such as increasing government spending or reducing taxes, which directly increases aggregate or effective demand in the economy.

(A) is a supply-side policy, not a Keynesian demand-side policy.

(B) is a microeconomic or supply-side approach related to market efficiency and provision of goods, not directly a Keynesian measure for cyclical unemployment.

(C) relates to production methods and capital intensity, which is a microeconomic or long-term structural consideration, not a Keynesian demand-management approach

17-D:

The diagram illustrates the liquidity preference (LP) curve, which shows the relationship between the rate of interest and the quantity of money demanded. The liquidity trap occurs at very low interest rates where the demand for money becomes perfectly elastic (horizontal).

At point D, the interest rate is at its lowest level shown in the diagram. At this point, the liquidity preference curve is horizontal, which means that any increase in the quantity of money supplied (a shift to the right along the horizontal segment) will not lead to a further decrease in the interest rate. People prefer to hold all their assets as liquid cash rather than invest in bonds because interest rates are already so low, rendering traditional monetary policy ineffective.

18-B:

An increase in government spending on education is intended to improve the skills and qualifications of the workforce. This leads to an increase in human capital, which allows workers to move more easily between different jobs, occupations, or industries, thereby increasing occupational mobility.

(A) is incorrect because better education reduces the supply of unskilled labor and increases the supply of skilled labor.

(C) is incorrect because increased skills generally improve employment prospects and reduce the natural rate of unemployment.

(D) is incorrect because while education can impact the labor market, it does not directly or necessarily lead to an increase in trade union power.

19-C:

A worsened trade balance means that imports are greater than exports, leading to a deficit. With a fixed exchange rate, the government or central bank must intervene to maintain the value of the currency. This typically involves using foreign currency reserves to buy back its own currency, or increasing interest rates to attract foreign capital.

The level of unemployment: The measures taken to defend the currency, such as increasing interest rates, can slow down the domestic economy, leading to a decrease in aggregate demand and an increase in unemployment.

The price level: The decrease in aggregate demand puts downward pressure on prices, leading to a decrease in the price level (disinflation or deflation).

20-C:

In economics, the long-run aggregate supply (LRAS) curve is vertical at the level of potential GDP (or full employment output). Operating on this vertical section means the economy is already at full capacity, utilizing all available resources efficiently. Therefore, attempts by the government to stimulate further growth of real GDP (e.g., through expansionary fiscal or monetary policy) will only lead to inflation, as output cannot increase beyond this point in the long run. The other options are incorrect because:

A small multiplier (A) would make government stimulus less effective but wouldn't prevent any growth if there's spare capacity.

A low consumer price index (B) suggests a lack of inflationary pressure, indicating there is likely room for growth.

A large negative output gap (D) means the economy is operating below full capacity, suggesting that government intervention could effectively achieve a faster rate of growth.

21-C:

A fall in domestic investment leads to an increase in unemployment. When unemployment rises, more people claim unemployment benefits, which increases government spending. Simultaneously, fewer people are earning income and paying income taxes, and reduced economic activity means lower corporate tax receipts. This combination of increased spending and decreased tax revenue leads to an increase in the government budget deficit.

A- An increase in interest rates on loans for house purchases: This is incorrect. In a recessionary environment with falling demand, central banks typically lower interest rates to stimulate borrowing, investment, and consumption.

B- an increase in the current account balance of payments deficit: This is incorrect. A fall in domestic demand usually leads to a decrease in demand for imports. This would improve the current account balance (reduce the deficit or increase the surplus).

D- an increase in the rate of price inflation: This is incorrect. A fall in aggregate demand creates spare capacity in the economy and puts downward pressure on prices, leading to lower inflation or deflation.

22-A:

The government's measures involve increasing government spending (budget deficit) and increasing the money supply, both of which aim to stimulate aggregate demand and overcome deflation. The effectiveness of these measures relies on people and businesses spending the increased money supply.

A- A high marginal propensity to save means that a large portion of any additional income is saved rather than spent. This reduces the multiplier effect of government spending and makes the expansionary policy less effective.

B- A rise in business confidence would encourage investment, which would support the government's measures, not reduce their effectiveness.

C- An inelastic demand for money means that interest rate changes have a large effect on the quantity of money demanded, which is generally not relevant to the effectiveness of increasing the money supply in this context.

D- Low cash deposit ratios for commercial banks would allow banks to create more credit from any given increase in reserves, which would enhance the effectiveness of increasing the money supply, not reduce interest rates.

23-D:

A revaluation of the exchange rate affects the economy in the short run (SR) and long run (LR) through various channels, impacting both demand-pull and cost-push inflation. The Marshall-Lerner rule ensures that the revaluation worsens the balance of trade over time. In the short run, a currency revaluation makes imports cheaper and exports more expensive.

Impact on Aggregate Demand (AD): Assuming the Marshall-Lerner condition holds, the initial effect on net exports is a decrease, shifting the AD curve to the left (e.g., from AD₀ to AD₁). This is because the volume of exports falls and the volume of imports rises, reducing overall demand for domestically produced goods and services. This movement would be towards point C or D from E on the diagram.

Impact on Aggregate Supply (AS): Cheaper imports, especially raw materials and intermediate goods, reduce the costs of production for firms. This causes the short-run aggregate supply (SRAS) curve to shift to the right (e.g., from AS₀ to AS₂). This effect mitigates cost-push inflation pressure.

Inflation Effects: The leftward shift in AD exerts downward pressure on demand-pull inflation, while the rightward shift in SRAS reduces cost-push inflation. The new short-run equilibrium is likely at a lower price level and potentially lower national income that is on option D.

24-B:

The Kuznets curve is a hypothetical curve that suggests that as an economy develops from a low-income agricultural society to a high-income industrial society, market forces first increase and then decrease economic inequality. (see the diagram below)

Option B is correct because the theory says that in the early stages of industrialization, investment in physical capital (factories, infrastructure) by a small segment of the population drives growth, causing the Gini coefficient (a measure of inequality) to rise as urban incomes increase relative to rural incomes. (showing a positive relationship between inequality and rising GDP per capita)

Option A is incorrect because the Gini coefficient is theorized to initially rise, not fall, during the early stages of development. It only begins to fall after a certain threshold of income is reached.

Options C and D are incorrect because the relationship between GDP per capita and the Gini coefficient according to the Kuznets curve is not always linear (either negative or positive). It is an inverted U-shaped relationship, meaning it is positive initially and then becomes negative later in the development process.

25-A:

The balance of payments is an accounting record of all monetary transactions between a country and the rest of the world. It is divided into the current account, capital account, and financial account.

The current account primarily records transactions in goods, services, investment income (like interest and dividends), and current transfers (remittances).

Option B is incorrect because portfolio investment is part of the financial account, not the current account.

Options C and D are incorrect because investment income and remittances are part of the current account, while the financial account deals with transactions in financial and fixed assets.

26-B:

A multinational corporation (MNC) is defined as a company that operates in two or more countries. Option B accurately describes this, as merely trading internationally (Option D) does not necessarily mean the firm has physical operations or facilities in those other countries. The other options are incorrect descriptions of an MNC's fundamental nature.

27-A:

A monetary union is an agreement between two or more states to share a single currency and a common monetary policy.

(A) Countries have the same currency: This is the defining characteristic of a monetary union, such as the Eurozone.

(B) Countries use the same fiscal policy: Fiscal policy (government spending and taxation) is typically set independently by each national government within a monetary union, although there may be coordination or rules (e.g., the EU's Stability and Growth Pact).

(C) Countries have the same domestic rates of sales tax: Tax rates, including sales tax (VAT), vary between countries in a monetary union.

(D) The government budget in each country is balanced: This is not a requirement; countries within a monetary union often run budget deficits or surpluses.

28-D:

The World Bank is an international financial institution that provides loans and grants to the governments of low- and middle-income countries for the purpose of pursuing capital projects. Its primary goal is to reduce poverty and improve living standards by promoting sustainable development.

Option A describes the role of the International Monetary Fund (IMF).

Option B is related to the World Trade Organization (WTO).

Option C describes a role of the IMF, which offers short-term assistance for balance of payments problems.

Option D accurately describes the core function of the World Bank.

29-A:

A rise in the domestic interest rate in an economy with a fixed exchange rate has the following impacts:

Financial Account: Higher domestic interest rates attract capital inflows from foreign investors seeking better returns (hot money). This inflow of foreign currency improves the financial account balance, leading to a surplus.

Current Account: Higher interest rates generally reduce domestic aggregate demand by increasing the cost of borrowing for consumption and investment. Lower domestic demand typically leads to a decrease in imports. This reduction in imports improves the current account balance.

Overall Impact: Both the current account and the financial account improve, matching option A.

30-C:

Balance of Payments Deficit: Greece has a persistent deficit, meaning the value of its imports is greater than the value of its exports.

Floating Exchange Rate: If Greece were to switch to a floating exchange rate, the value of its currency (if not tied to the euro) would fall or "depreciate" due to the high demand for foreign currency to pay for imports and the low demand for its own currency.

Impact of Depreciation: A weaker currency makes a country's exports cheaper for foreigners to buy, and imports more expensive for domestic consumers. This increases export competitiveness and reduces import demand, which helps to correct a balance of payments deficit.

Why other options are incorrect:

- * (A) Floating exchange rates are typically more volatile than fixed ones.
- * (B) Floating exchange rates are generally more open to speculation.
- * (D) While a floating rate helps correct the balance, it does not automatically or perfectly balance the value of exports and imports.